SHASHIDHER MYLARAPU

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PROFESSIONAL SUMMARY

- AI/ML Engineer with 4+ years of combined experience developing Machine Learning, Deep Learning, and Generative AI solutions across automotive manufacturing, enterprise AI, and cloud-based environments.
- Experienced in building predictive models, feature engineering pipelines, and NLP applications using Python, Scikit-learn, XGBoost, PyTorch, TensorFlow, and Hugging Face Transformers to address business and operational challenges.
- Skilled in developing Generative AI and RAG applications using LangChain, LangGraph, Amazon Bedrock, and Vector Databases, with exposure to ML deployment, MLOps, monitoring, and CI/CD using AWS SageMaker, FastAPI, Docker, and MLflow.
- Connects AI/ML engineering with manufacturing quality, document intelligence, and operational needs, collaborating with engineering, QA, and business teams to improve prediction reliability, processing efficiency, and model performance.

TECHNICAL SKILLS

Programming & AI Development: Python, SQL, FastAPI, REST APIs, Git, Linux, Pytest

Machine Learning & Algorithms: Scikit-learn, XGBoost, LightGBM, Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Hyperparameter Tuning, Cross-Validation

Deep Learning & NLP: PyTorch, TensorFlow, Hugging Face Transformers, Neural Networks, Natural Language Processing (NLP), Transfer Learning, Text Classification

Generative AI & Agentic AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Databases, FAISS, Semantic Search, Hybrid Search, Reranking, AI Agents, Tool Calling, Structured Outputs

MLOps & Model Lifecycle: MLflow, Experiment Tracking, Model Registry, Model Versioning, Model Deployment, Model Serving, CI/CD for ML, Model Monitoring, Data Drift, Model Drift

Cloud AI & Deployment: AWS, Amazon SageMaker, Amazon Bedrock, Amazon S3, Docker, Kubernetes, GitHub Actions

AI Data Processing & Inference: NumPy, Pandas, PySpark, Data Preprocessing, Feature Pipelines, Batch Inference, Real-Time Inference

Model Evaluation & Responsible AI: Precision, Recall, F1 Score, ROC-AUC, LLM Evaluation, Hallucination Detection, AI Guardrails, SHAP, Model Explainability

PROFESSIONAL EXPERIENCE

AI/ML Engineer **May 2026 - Present**

Ford Motor Company | USA | Remote

- Engineered manufacturing defect prediction models using Python, XGBoost, Scikit-learn, and Feature Engineering, improving classification F1 Score by 8% to support earlier detection of production quality risks.
- Formulated production-risk forecasting models with LightGBM, Pandas, and Cross-Validation to identify abnormal manufacturing patterns, helping quality engineers prioritize investigations before recurring defects affect operations.
- Applied Retrieval-Augmented Generation (RAG) with Amazon Bedrock, LangChain, Embeddings, and Vector Databases, reducing engineering document retrieval time by 24% during internal pilot evaluations.
- Orchestrated Agentic AI workflows using LangGraph, Tool Calling, and Structured Outputs to automate engineering knowledge requests, coordinate dependent actions, and reduce manual navigation across technical resources.
- Deployed ML inference services through AWS SageMaker, Docker, FastAPI, and REST APIs, maintaining 99.5% availability for internal manufacturing analytics applications.
- Optimized real-time inference endpoints with Python, FastAPI, and Batch Inference, reducing prediction latency by 19% to deliver faster model outputs for operational decision-making.
- Established MLOps workflows using MLflow, Model Registry, GitHub Actions, and CI/CD for ML, shortening model promotion cycles by 21% through automated validation and repeatable deployment procedures.
- Investigated model degradation using SHAP, Model Monitoring, Data Drift, and Model Drift analysis, partnering with manufacturing engineering and QA teams to strengthen prediction reliability and investigate production exceptions.

AI/ML Engineer **Aug 2021 - Dec 2024**

Augur Talentcare Pvt Ltd | India | Remote

Client Domain: AI / Cloud / Enterprise

Project: Enterprise Document Intelligence & Cloud-Based AI Automation

- Architected enterprise document classification workflows using Python, PyTorch, Hugging Face Transformers, and NLP to process unstructured business records, achieving 0.89 macro F1 across document categories.
- Built deep learning pipelines with TensorFlow, Transfer Learning, and Text Classification to extract document context and classify incoming records, reducing manual categorization effort by 32%.
- Integrated machine learning inference APIs using FastAPI, REST APIs, and Docker, enabling enterprise applications to submit documents and receive structured classification results without manual processing.
- Designed cloud-based model training workflows using AWS SageMaker, Amazon S3, and MLflow, reducing model experimentation turnaround by 27% through repeatable training and evaluation processes.
- Implemented NLP preprocessing and embedding-based similarity techniques using Hugging Face Transformers and Python to identify related document content, supporting consistent classification of previously unseen records.
- Automated model validation and release workflows using GitHub Actions, Pytest, and Model Registry, reducing manual release preparation by 38% while strengthening deployment consistency.
- Accelerated inference performance through Python, FastAPI, and model-serving improvements, supporting 18,000 daily document predictions for enterprise application workflows.
- Evaluated classification reliability using Precision, Recall, F1 Score, and SHAP, collaborating with application engineering and operations teams to investigate misclassified records and refine acceptance criteria.

Machine Learning Intern **Aug 2020 - Jul 2021**
Augur Talentcare Pvt Ltd | India | Remote
Client Domain: AI / Cloud / Enterprise
Project: Customer Behavior Prediction & Machine Learning Analytics

- Prepared customer behavior datasets using Python, Pandas, and NumPy, cleaning 85,000 records to resolve missing values, duplicate observations, and inconsistent features affecting model training.
- Analyzed historical customer activity using Python, SQL, and exploratory analysis, identifying 14 predictive features associated with engagement patterns and customer response behavior.
- Created baseline customer propensity models using Scikit-learn and Supervised Learning techniques, achieving 0.81 F1 Score on held-out evaluation datasets.
- Experimented with classification algorithms and Cross-Validation to compare customer response predictions, investigate model errors, and recommend suitable approaches for further development.
- Refined feature selection and Hyperparameter Tuning using Scikit-learn, improving validation ROC-AUC by 0.05 through iterative experiments and supervised model comparisons.
- Documented ML experiments using Python, Git, and Pytest, collaborating with senior engineers to review model assumptions, reproduce results, and maintain consistent evaluation practices.

PROJECTS

Enterprise Knowledge Assistant with Agentic RAG

- Created an enterprise knowledge assistant using Amazon Bedrock, LangChain, RAG, and FAISS, targeting 89% grounded-answer accuracy for questions requiring information from internal knowledge documents.
- Coordinated multi-step retrieval workflows using LangGraph, Prompt Engineering, and Tool Calling across 6 knowledge-request scenarios, enabling context-aware responses and structured outputs.
- Assessed response quality using LLM Evaluation, Hallucination Detection, and AI Guardrails, targeting a 22% reduction in unsupported responses before exposing the assistant through FastAPI.

Intelligent Visual Defect Detection & Quality Inspection

- Developed a deep learning image classification model using PyTorch and Transfer Learning, targeting 0.92 F1 Score to distinguish manufacturing surface defects from acceptable product conditions.
- Calibrated model training with TensorFlow, Hyperparameter Tuning, and Cross-Validation across 14,000 labeled images, improving classification consistency under varied inspection conditions.
- Validated model experiments using AWS SageMaker and MLflow, aiming to reduce manual inspection review effort by 26% through confidence-based classification and exception handling.

Predictive Customer Churn & Retention Modeling

- Constructed a customer churn prediction model using Python, XGBoost, and Scikit-learn, targeting 0.86 ROC-AUC to identify subscribers at risk of leaving and support retention planning.
- Derived behavioral features from 72,000 customer records using SQL, Pandas, and NumPy, capturing engagement frequency, service usage, and renewal patterns for predictive modeling.
- Quantified customer risk through Cross-Validation and classification metrics, defining 5 actionable retention segments to help business teams prioritize targeted outreach.

EDUCATION

Master of Science in Information Technology **Jan 2025 - Aug 2026**
Southern New Hampshire University | USA

Bachelor of Technology in Electronics and Communication Engineering **Aug 2019 - Aug 2023**
Jawaharlal Nehru Technological University | India

CERTIFICATIONS

- IBM AI Engineering Professional Certificate - **Coursera**
- IBM Machine Learning Professional Certificate - **Coursera**
- Deep Learning Specialization - **Coursera**
- IBM Generative AI Engineering Professional Certificate - **Coursera**
- Machine Learning Specialization - **Coursera**
- Machine Learning in Production - **Coursera**